

Healthcare resource utilization and associated cost in patients with metastatic non-small cell lung cancer treated in the immunotherapy era

Lior Apter^{1,2,*}, Sarah Sharman Moser^{3,†}, Sivan Gazit³, Gabriel Chodick^{3,4}, Moshe Hoshen³, Dan Greenberg^{2,†}, Nava Siegelmann-Danieli^{1,4,†}

¹Health Division, Maccabi Healthcare Services, Tel Aviv 6812509, Israel,

²Department of Health Policy and Management, School of Public Health, Faculty of Health Sciences, Ben-Gurion University of the Negev, Be'er-Sheva 8410501, Israel,

³Maccabi Institute for Research and Innovation (Maccabitech), Maccabi Healthcare Services, Tel Aviv 6812509, Israel,

⁴Faculty of Medical and Health Sciences, Tel Aviv University, Tel Aviv 6997801, Israel

*Corresponding author: Lior Apter, Health Division, Maccabi Healthcare Services, 27 Hamered St, Tel Aviv 6812509, Israel (apter_li@mac.org.il).

†First two and last two authors had equal contributions.

Abstract

Background: Treatment approach for metastatic non-small cell lung cancer (mNSCLC) has revolutionized in the recent decade with the introduction of immunotherapy and targeted medications in first-line (1L) therapy. We present real-world data on clinical outcomes and direct healthcare resource utilization (HCRU) and cost in a 2.7-million-member Israeli health provider.

Patients and methods: Newly diagnosed mNSCLC patients between January 2017 and December 2020 were categorized by 1L treatment: platinum-based chemotherapy, targeted therapy, or immunotherapy. HCRU and costs were calculated based on the Ministry of Health Prices and were assessed at a minimum of 6 months' follow-up (cutoff: 30 June 2021).

Results: A total of 886 patients were included in the study: 40.6% female, median age 68 years (IQR 61–74), 24.3% never smokers, 80.6% with adenocarcinoma, and 54% with a 0–1 performance status. The median follow-up was 27.12 months (95% CI, 24.7–29.6) and the median duration of first-line (1L) treatment was 2.3 months for platinum-based chemotherapy ($n = 177$), 12.3 months for targeted therapy ($n = 255$), and 4.8 months for immunotherapy ($n = 463$). The median overall survival was 9.09, 27.68, and 12.46 months, respectively. Total 1L costs were driven by radiotherapy for platinum-based chemotherapy and medication for targeted therapy or immunotherapy. Total costs for deceased patients over the entire follow-up were €121 155, €129 458, and €110 716, respectively.

Conclusion: The treatment of mNSCLC carries a high economic burden, primarily driven by first-line therapy, especially with targeted and immune therapies. Further studies are needed to evaluate the impact of innovative treatments on the disease management costs of mNSCLC.

Key words: non-small cell lung cancer; HCRU.

Implications for practice

Treatment approach for metastatic non-small cell cancer (mNSCLC) has revolutionized in the recent decade with the introduction of first-line (1L) immunotherapy and targeted therapy. Despite improvement in treatment options, a high economic burden associated with the treatment of mNSCLC exists, driven mainly by systemic 1L therapy, which may reflect the greater use of targeted and immune therapies. Further studies are required to assess the impact of immunotherapy and other innovative treatments on the disease-management costs of mNSCLC.

Introduction

Lung cancer, the second most prevalent cancer globally, is the primary cause of cancer-related deaths worldwide.^{1–3} Non-small cell lung cancer (NSCLC) constitutes approximately 80%–90% of all lung cancer cases.^{2,3} The five-year survival rate for patients with metastatic NSCLC (mNSCLC) was reported as only 6%–7% in a comprehensive patient database spanning the years 1999–2010. However, a notable

improvement in survival rates has been observed from 2014 to 2018, attributed to factors such as reduced smoking rates and advancements in treatment options.^{2,4,5} Guidelines for systemic anti-cancer therapy in mNSCLC vary based on tumor histology and the presence of actionable driver mutations suitable for personalized treatment.⁶ The landscape of treatment has rapidly progressed in recent years, with clinical recommendations advocating for targeted therapy using tyrosine

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kinase inhibitors in patients testing positive for driver mutations like epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase, or proto-oncogene tyrosine-protein kinase ROS enzyme (ROS1). For patients testing negative for driver mutations, there has been a paradigm shift in the standard of care, transitioning from platinum-based doublet chemotherapy regimens to immune checkpoint inhibitors.

The initial approval for a single-agent immunotherapy targeting the programmed death-1/programmed death-ligand1 (PD-1/PD-L1) inhibitor in metastatic non-small cell lung cancer (mNSCLC) came with nivolumab for second-line treatment (2L). This approval by the Food and Drug Administration (FDA) was granted on March 4, 2015, and public reimbursement in Israel began in January 2016.⁷ Subsequently, based on findings from the Keynote-024 trial, pembrolizumab received FDA approval on October 24, 2016, for first-line (1L) treatment as a single-agent therapy in mNSCLC cases characterized by a PD-L1 tumor proportion score (TPS) of $\geq 50\%$ and the absence of actionable mutations (EGFR/ALK).^{8,9}

As of January 2017, public reimbursement for 1L pembrolizumab monotherapy has been in effect in Israel. Additionally, in January 2019, approval was granted for the use of pembrolizumab in combination with chemotherapy as a 1L treatment for advanced NSCLC in Israel, regardless of PD-L1 TPS level, based on the findings of the Keynote-189 trial.¹⁰⁻¹⁷

Considering the transformation in the treatment landscape in Israel, the economic burden associated with NSCLC may have changed. Before the immunotherapy era, studies have shown that NSCLC was associated with a significant economic burden, and healthcare resource utilization (HCRU) was primarily driven by hospitalizations for adverse event (AE) management.¹⁸⁻²¹ Other studies examining the impact of newer systemic treatments on HCRU reported increased use of ambulatory cancer services with decreased hospital admissions. These studies were of relatively small sample size, had short follow-up periods, and focused mainly on patients receiving targeted therapy.²²⁻²⁷

Israel implemented a universal healthcare system in 1995 through the National Health Insurance Law, with a standard list of healthcare services that all residents are entitled to receive via four national health funds in the community and public hospitals. The national list of reimbursed services is referred to as the National Formulary or the Basket of Health Services. This system operates with a low co-payment structure, and oncology drugs are provided without any co-payment requirement. Additionally, Israel utilizes a centralized approach to health technology assessment. A public committee, appointed annually by the Ministers of Health and Finance, known as The National Advisory Committee for the Basket of Health Services, makes decisions regarding reimbursed technologies. This committee recommends new and innovative technologies and drugs to be included in the standard national formulary list, which is covered by the health funds in the subsequent year, within a pre-allocated budget. Budget impact assessments are conducted for all potential new treatments, comparing their added value to existing technologies in the national formulary. The committee makes final evaluations and decisions based on clinical, social, and ethical considerations between October and December each year. Subsequently, it advises the Ministry of Health on which new technologies should be integrated into the national health insurance, effective from January of the following year.²⁸

The aim of this study was to describe the economic burden for all patients in a real-world cohort of patients that initiated 1L treatment for mNSCLC between January 2017 and December 2020. We also describe baseline demographic characteristics, time on treatment (ToT), and overall survival (OS) for 1L therapy. To the best of our knowledge, we present the only real-world study on HCRU and costs in Israel, where reimbursed drugs are publicly covered.

Materials and methods

Data source

This retrospective cohort study was conducted on anonymized records derived from the computerized databases of Maccabi Healthcare Services (MHS), a nationwide healthcare insurer-provider. MHS has approximately 2.7-million-members, constituting over a quarter of the Israeli population, and shares sociodemographic characteristics similar to those of the general population.²⁹ The MHS database contains longitudinal data automatically collected since 1993 for a stable population (with less than 1% of members annually transitioning to other health plans), encompassing laboratory results from a single central laboratory, pharmacy prescriptions and purchasing data, hospitalizations, procedures, and consultations, drug approval requests to the MHS Medication Approval Committee (which issues pre-authorization for cancer treatment), and actual treatment days. MHS uses the *International Classification of Diseases, Ninth Revision, Clinical Modification* (ICD-9-CM) coding systems, as well as self-developed coding systems to provide more detailed diagnostic information. Procedures are coded using *Current Procedural Terminology* codes.

Moreover, additional information was manually abstracted from de-identified patients' hospital reports or MHS records including disease staging, histological details, metastasis's location, Eastern Cooperative Oncology Group (ECOG) performance status (PS), and genomic tumor driver mutations.

Study population

We identified members with a confirmed mNSCLC in the MHS cancer registry, diagnosed between January 1, 2017 and December 31, 2020. The MHS cancer registry is compiled from information from the National Cancer Registry and pathology in cases diagnosed in the MHS central pathology lab, and cancer treatment approvals by the MHS Medication Approval Committee.

Patients were included in this cohort if they were at least 18 years of age at diagnosis, had at least 1-year of continuous enrollment in MHS before diagnosis date, and initiated systemic 1L treatment.

Patients were treated until disease progression or serious adverse effect and up to 24 months for those receiving immunotherapy as per NCCN guidelines.

Data were collected until June 30, 2021 to allow for at least 6 months of follow-up. Index date was set as the date of 1L treatment initiation.

Study variables

Demographic and clinical data collected included age at index date, sex, socioeconomic status, district, prevalence of comorbid conditions, body mass index (BMI), and smoking status. Socioeconomic status was categorized into quartiles based on the poverty index of the member's enumeration area

at the neighborhood level.^{30,31} The poverty index is based on several parameters including household income, educational level, crowding, physical conditions, and car ownership. BMI was defined as the closest to index date within 5-year period before index date. Smoking data were collected from physician reporting and classified into ever, never, or unknown.

Co-morbidities at baseline were identified using MHS registries (diabetes mellitus, cardiovascular disease, hypertension, osteoporosis, and chronic obstructive pulmonary disease).³²⁻³⁴ These registries are continuously updated, and identify patients via automatic search formulas.³²⁻³⁴

The Deyo-Charlson co-morbidity index (CCI), using ICD-9-CM codes and MHS registries to determine presence/absence of disease, was calculated.³⁵ Imaging (X-ray, CT, PET-CT, MRI) results, biopsy results, metastasis's location, driver mutation testing and PD-L1 testing were collected manually. PD-L1 testing results were categorized as TPS $\geq$ 50% or TPS < 50%. Patients who tested negative for EGFR/ALK/ROS1 driver mutations or untested were defined as negative/unknown. ECOG PS was based on physician reporting in the medical records at baseline.

Measures of HCRU and costs

HCRU and costs were calculated for primary care physician visits, emergency department (ED) visits, imaging, radiotherapy, cancer medications, and all prescription drugs purchased and inpatient hospital admissions.

Costs per-patient-per-month (PPPM) for all services except drug costs were reported as total costs based on the Israeli Ministry of Health Price List for Ambulatory and Hospitalization Services published on May 01, 2023. Pharmaceutical prices are based on the Israeli Ministry of Health prescription drug price list, published on January 01, 2023. These lists indicate the maximum list price for health-care services and pharmaceuticals. The exchange rate shekel to Euro at the time of calculation was 3.88. In Israel, oncological patients are exempt from medication co-payment.

HCRU and costs were calculated PPPM for 3 different periods: within 1L drug treatment period, during the entire follow-up, and within 1-month before death. The total cost for the follow-up period was calculated for those patients who had already died.

The distribution of treatments was categorized by class: platinum-based chemotherapy, targeted therapies (TKI), and immunotherapy (PD-1/PD-L1 inhibitor therapy).

Data analysis

Statistical analysis

Descriptive analyses were conducted to evaluate the demographic and clinical characteristics of the study cohort by histology. Categorical variables were reported as frequency and percentage and the association investigated using chi-square testing, and continuous variables were reported as median (interquartile range [IQR]) and compared using the non-parametric Kruskal-Wallis test or mean (standard deviation) and compared with the *t*-test. Time to event analysis for time on treatment (ToT) and OS were assessed using Kaplan-Meier analysis; median time to event was reported with 95% confidence intervals (CI). For ToT analysis, individuals were followed from index date until the outcome (discontinuation of treatment), death, loss to follow-up, or end of study period (June 30, 2021), which ever occurred first. Patients

were considered discontinued if their last treatment purchase was more than 120 days prior to their follow-up date or they switched to second-line treatment. OS was assessed starting from index using the all-cause mortality data from the National Insurance Institute. Individuals were followed from index date until death, loss to follow-up, or end of study period (June 30, 2021), whichever occurred first. All analyses were presented for 1L treatment.

All analyses were conducted using IBM SPSS Statistics for Windows, Version 22.0. Armonk, NY: IBM Corp and R version 3.5.1, and a *P* value < .05 was considered statistically significant for all analyses.

The study was approved by the local ethics review board of MHS in Israel (approval number 0077-23-MHS).

Results

Patient characteristics and clinical outcomes

This cohort consisted of 886 patients with histologically confirmed mNSCLC who initiated 1L treatment within the study period (2017–2020, inclusive); 177 (20%) received platinum-based chemotherapy, 225 (25%) received targeted therapy, and 463 (52%) received immunotherapy (21 [<3%] received unknown or other treatment). Median age at start of treatment was 68 years (IQR 61-74), 40.6% were female, 75.7% were confirmed smokers, 20.3% had brain metastases, and 53.7% with ECOG PS = 0-1 (missing 27%) (Table 1). Median follow-up for the entire cohort was 27.12 months (95% CI, 24.66-29.59).

Patients with an actionable mutation received targeted therapy. In more than 97% of patients that received immunotherapy drugs (monotherapy or in combination with chemotherapy), the treatment agent was pembrolizumab.

The group of patients receiving targeted therapy for driver mutations had a higher proportion of female patients (59.1%) as compared with the platinum-based chemotherapy (32.2%) or immunotherapy (34.3%) groups; they were also more likely to be never smokers (60%) versus the other groups, (18.6% and 9.5%, respectively), and have predominantly adenocarcinoma histology (95.6%), as compared with the other treatment groups (76.3% and 75.6%, respectively). In addition, patients who received targeted therapy had a higher proportion of brain metastases at baseline (29.8%) than those who received platinum-based chemotherapy (9.6%) or immunotherapy (19.9%).

Median ToT on 1L for patients treated with platinum-based chemotherapy, targeted therapy, and immunotherapy was 2.3, 12.3, and 4.8 months, respectively (Table 2).

Median OS for patients treated in 1L with platinum-based chemotherapy, targeted therapy, and immunotherapy was 9.09, 27.68, and 12.46 months, respectively (Table 2).

HCRU outcomes

During the 1L period

During the entire 1L treatment period, patients receiving targeted therapy, as compared with platinum-based chemotherapy and immunotherapy inhibitors groups, had a higher proportion of primary care visits (95.1% vs 90.4% and 90.1%, *P* = .012), ED visits (36.9% vs 26.6% and 32.8%, *P* = 0.019) and imaging studies performed (85.3% vs 59.9% and 71.7%, *P* < .001), respectively. When corrected for treatment duration, median (IQR) number of physician visits

Table 1. Baseline characteristics of patients diagnosed with metastatic NSCLC during 2017-2020 and initiation 1L treatment, in the year before treatment initiation, $n = 886$. All results n (%) except where stated.

	Overall*, $N = 886$	Platinum-based chemotherapy, $N = 177$ (19.9%)	Targeted therapy, $N = 225$ (25.4%)	Immunotherapy, $N = 463$ (52.3%)	P-value
Age (median [IQR])	68.00 [61.00, 74.00]	67.00 [62.00, 75.00]	66.00 [58.00, 74.00]	68.00 [62.00, 74.00]	.444
18-34	3 (0.3)	0 (0.0)	2 (0.9)	1 (0.2)	.043
35-64	333 (37.6)	68 (38.4)	91 (40.4)	165 (35.6)	
65-74	336 (37.9)	62 (35.0)	78 (34.7)	194 (41.9)	
75+	214 (24.2)	47 (26.6)	54 (24.0)	103 (22.2)	
Female sex	360 (40.6)	57 (32.2)	133 (59.1)	159 (34.3)	<.001
District					
Center	563 (63.5)	110 (62.1)	142 (63.1)	298 (64.4)	.502
North	166 (18.7)	33 (18.6)	35 (15.6)	94 (20.3)	
South	157 (17.7)	34 (19.2)	48 (21.3)	71 (15.3)	
Socioeconomic status					
Low	331 (37.4)	86 (48.6)	84 (37.3)	154 (33.3)	.033
Medium	186 (21.0)	31 (17.5)	44 (19.6)	107 (23.1)	
High	369 (41.6)	60 (33.9)	97 (43.1)	202 (43.6)	
Deyo-charlson co-morbidity index	3.95 (3.35)	3.89 (3.49)	3.72 (3.37)	4.06 (3.29)	.517
0	171 (19.3)	33 (18.6)	51 (22.7)	85 (18.4)	.399
1-2	228 (25.7)	52 (29.4)	55 (24.4)	115 (24.8)	
3-6	258 (29.1)	48 (27.1)	73 (32.4)	131 (28.3)	
7+	229 (25.8)	44 (24.9)	46 (20.4)	132 (28.5)	
Diabetes mellitus	235 (26.5)	49 (27.7)	54 (24.0)	127 (27.4)	.769
Cardiovascular disease	281 (31.7)	70 (39.5)	48 (21.3)	156 (33.7)	.001
Hypertension	451 (50.9)	98 (55.4)	116 (51.6)	224 (48.4)	.302
Chronic obstructive pulmonary disease	172 (19.4)	44 (24.9)	12 (5.3)	111 (24.0)	<.001
Osteoporosis	230 (26.0)	44 (24.9)	63 (28.0)	116 (25.1)	.708
Never smokers	215 (24.3)	33 (18.6)	135 (60.0)	44 (9.5)	<.001
Body mass index (median [IQR])	25.74 [22.91, 29.38]	25.46 [23.12, 30.47]	26.68 [24.00, 30.03]	25.50 [22.43, 28.77]	.007
Histology					
Adenocarcinoma	714 (80.6)	135 (76.3)	215 (95.6)	350 (75.6)	<.001
Squamous cell carcinoma	129 (14.6)	34 (19.2)	3 (1.3)	89 (19.2)	
Other	43 (4.9)	8 (4.5)	7 (3.1)	24 (5.2)	
ECOG performance status					
0-1	476 (53.7)	69 (39.0)	140 (62.2)	259 (55.9)	<.001
2	116 (13.1)	31 (17.5)	19 (8.4)	64 (13.8)	
3-4	55 (6.2)	5 (2.8)	12 (5.3)	36 (7.8)	
Missing	239 (27.0)	72 (40.7)	54 (24.0)	104 (22.5)	
Brain metastases	180 (20.3)	17 (9.6)	67 (29.8)	92 (19.9)	<.001
Liver metastases	166 (18.7)	31 (17.5)	45 (20.0)	87 (18.8)	.877
Bone metastases	377 (42.6)	77 (43.5)	112 (49.8)	177 (38.2)	.026
PD-L1 tested	752 (84.9)	139 (78.5)	154 (68.4)	445 (96.1)	<.001
<50	431 (48.6)	131 (74.0)	96 (42.7)	191 (41.3)	<0.001
>50	316 (35.7)	7 (4.0)	56 (24.9)	252 (54.4)	
Missing	139 (15.7)	39 (22.0)	73 (32.4)	20 (4.3)	

*Including patients that received any other treatment $n = 21$ (or < 3%).

Abbreviations: ECOG, Eastern Cooperative Oncology Group; IQR, inter-quartile range; NSCLC, non-small cell lung cancer; PD-L1, programmed death-ligand 1.

Table 2. Time on treatment (ToT) and overall survival, by first-line treatment type, months, $n = 886$.

Treatment type*	<i>n</i>	Number (%) discontinued	Median time on treatment (95% CI)	Number (%) died	Median overall survival (95% CI), months
Platinum-based chemotherapy group	177	173 (97.74%)	2.27 (1.94, 2.86)	147 (83.05%)	9.09 (7.13, 11.2)
Targeted therapy group	225	140 (62.22%)	12.26 (10.65, 17.13)	100 (44.44%)	27.68 (24.16, 37.5)
Immunotherapy group**	463	383 (82.72%)	4.77 (4.18, 5.79)	287 (61.99%)	12.46 (10.72, 15.4)

*Results for patients that received any other treatment $n = 21$ are not shown.

**Monotherapy and chemo-immunotherapy.

PPPM (per-patient-per-month) were 4.19 (2.39, 6.38), 3.53 (2.60, 5.30), and 4.09 (2.14, 6.35) for patients who received platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively. Hospital admission rates PPPM were 24.3%, 34.7%, and 40.8%, for those groups, respectively ($P = 0.001$), and mean (SD) lengths of stay PPPM was 0.81 (0.43, 1.47), 0.55 (0.23, 1.06), and 0.81 (0.23, 2.34), respectively ($P = 0.08$). Median (IQR) number of pharmacological groups PPPM was 4.49 (2.32, 8.37), 1.49 (0.85, 2.94), and 2.59 (1.26, 5.88) ($P < .001$), and median (IQR) number of packs PPPM was 21.29 (12.37, 45.34), 12.34 (7.22, 26.90), and 19.74 (11.52, 39.44) for patients receiving platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively ($P < .001$, Table 3). See Table S1 for drug utilization for oncological and non-oncological medications.

Total costs PPPM for the 1L period were the highest between the time periods measured, €28 309 €21 422, and €46 717 for patients receiving platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively (Figure 1). Total costs PPPM were driven by oncology drug costs: €8503, €18 644, and €38 459 and radiotherapy costs: €18 376, €1528, and €6728 for patients receiving platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively.

During the month before death

A total of 555 (62.6%) patients died during the study follow-up period: 83%, 44%, and 62% of those who received platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively.

Total cost PPPM in the last month of life was €11 209, €18 264, and €11 536 for patients that initially received 1L platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively, and were attributed mostly to hospitalization, oncology drugs, and radiotherapy. Hospitalization costs PPPM increased further in the month before death to €4360, €4247, €3579 for patients that received 1L platinum-based chemotherapy, targeted therapy, and immunotherapy, respectively (Figure 1).

Total cost for the entire follow-up period

Overall survival was 9.09 months (7.13-11.2), 27.68 (24.16-37.5), and 12.46 (10.72-15.4) for patients that received 1L platinum-based chemotherapy, targeted therapy, and immunotherapy respectively. Total mean cost for the entire follow-up period was €121 154, €129 458, and €110 717 respectively for those that died ($n = 555$, Table 4).

Discussion

This retrospective descriptive study describes baseline characteristics, treatment patterns, HCRU, and costs in a real-world

cohort of unselected patients with mNSCLC. Overall, we followed data on 886 patients with confirmed mNSCLC who initiated 1L treatment with either platinum-based chemotherapy, targeted therapy, or immunotherapy (the latter as monotherapy or chemo-immunotherapy).

Our cohort's baseline characteristics are typical of mNSCLC cohorts overall.³⁶⁻⁴⁴ In comparison to other groups of patients in the study, patients treated with targeted therapy were younger, with a higher proportion of women, less smokers, and patients had better performance status.

1L treatment adhered to clinical guidelines in the study period³ and drug reimbursement in the Israel National Formulary. Patients with actionable mutations in their tumors received targeted therapy. For patients without actionable mutations, those with TPS $\geq 50\%$ received immunotherapy as a single agent through the entire study period, or chemo-immunotherapy from 2019 to 2020. Those with tumors exhibiting TPS $< 50\%$ were candidates for platinum-based regimens during the years 2017-2018 and for chemo-immunotherapy during the years 2019-2020, according to the Israel National Formulary. Treatment choice in this case was left to the discretion of the treating physician.

Patients were treated until disease progression or serious adverse effect: those receiving immunotherapy were treated for up to 24 months or until disease progression or serious adverse effect. In accordance with our daily work at the MHS Medication Approval Committee, the major cause for treatment discontinuation in metastatic patients is disease progression and/or death and in this study, 40% progressed beyond 1L and switched to 2L.

HCRU and costs

HCRU differed between 1L groups. Over 90% of all patients had primary care physician visits during the 1L period, likely for prescription renewals and/or referrals for imaging. Patients in the 1L targeted group had a higher number of imaging though it spanned over the longer treatment period of 1L in this group and likely reflects treatment evaluations from time to time in patients with relatively prolonged treatment benefit. This group also had less hospitalizations possibly due to better performance status overall and lower rate of side effects to treatment as seen in other studies.⁴⁵ In the 1L platinum-based chemotherapy group, more patients required early radiotherapy treatments, possibly due to higher rates of palliative radiotherapy, sometimes also chemo-irradiation regimens.

In the last month before death (for those who already died) HCRU was high, with similar hospital admissions for all groups, while radiotherapy for palliation was used more in the targeted treatment group who did not receive this modality earlier.

Table 3. Healthcare utilization resource by treatment type, *n* = 886.

	Overall*, <i>n</i> = 886	Platinum based chemotherapy, <i>n</i> = 177 (19.9%)	Targeted therapy, <i>n</i> = 225 (25.4%)	Immunotherapy, <i>n</i> = 463 (52.3%)	<i>P</i> -value
Median [IQR] number of days during 1L treatment period	148.00 [47.25, 365.75]	69.00 [42.00, 148.00]	312.00 [155.00, 537.00]	143.00 [46.50, 362.00]	<.001
Median [IQR] number of days during follow-up period	337.50 [146.25, 672.00]	277.00 [123.00, 679.00]	496.00 [232.00, 807.00]	316.00 [125.00, 572.00]	<.001
Primary care physician visit					
During 1L period	807 (91.1)	160 (90.4)	214 (95.1)	417 (90.1)	.012
Median [IQR] visits per person per month	3.98 [2.30, 6.12]	4.19 [2.39, 6.38]	3.53 [2.60, 5.30]	4.09 [2.14, 6.35]	.570
During FU period	869 (98.1)	173 (97.7)	223 (99.1)	452 (97.6)	.512
Median [IQR] visits per person per month	4.26 [2.79, 6.13]	4.44 [3.09, 6.43]	3.86 [2.73, 5.45]	4.41 [2.72, 6.40]	.179
Last month before death	499 (89.9)	134 (90.5)	91 (91.0)	258 (89.9)	.501
Median [IQR] visits per person per month**	5.00 [2.00, 8.00]	5.00 [2.75, 8.00]	5.00 [2.00, 9.00]	5.00 [2.00, 8.00]	.119
ED visit					
During 1L period	284 (32.1)	47 (26.6)	83 (36.9)	152 (32.8)	.019
Median [IQR] visits per person per month	0.00 [0.00, 0.10]	0.00 [0.00, 0.05]	0.00 [0.00, 0.09]	0.00 [0.00, 0.12]	.154
During FU period	470 (53.0)	105 (59.3)	114 (50.7)	243 (52.5)	.154
Median [IQR] visits per person per month	0.03 [0.00, 0.16]	0.05 [0.00, 0.18]	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]	.098
Last month before death	113 (20.4)	30 (20.3)	23 (23.0)	58 (20.2)	.624
Median [IQR] visits per person per month**	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.59
Imaging***					
During 1L period	638 (72.0)	106 (59.9)	192 (85.3)	332 (71.7)	<.001
Median [IQR] visits per person per month	0.40 [0.00, 0.62]	0.37 [0.00, 0.61]	0.45 [0.28, 0.63]	0.40 [0.00, 0.62]	.006
During FU period	766 (86.5)	158 (89.3)	208 (92.4)	385 (83.2)	.001
Median [IQR] visits per person per month	0.43 [0.28, 0.61]	0.42 [0.29, 0.60]	0.48 [0.32, 0.63]	0.41 [0.25, 0.59]	.009
Last month before death	133 (24.0)	34 (23.0)	30 (30.0)	65 (22.6)	.471
Median [IQR] visits per person per month**	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 1.00]	0.00 [0.00, 0.00]	.373
Radiotherapy					
During 1L period	179 (20.2)	26 (14.7)	50 (22.2)	101 (21.8)	.105
Median [IQR] visits per person per month	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.209
During FU period	345 (38.9)	78 (44.1)	78 (34.7)	182 (39.3)	.263
Median [IQR] visits per person per month	0.00 [0.00, 0.59]	0.00 [0.00, 0.91]	0.00 [0.00, 0.35]	0.00 [0.00, 0.68]	.092
Last month before death	44 (7.9)	9 (6.1)	9 (9.0)	25 (8.7)	.723
Median [IQR] visits per person per month**	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.754

Table 3. Continued

Drug treatment	During 1L period		Overall*, n = 886		Platinum based chemotherapy, n = 177 (19.9%)		Targeted therapy, n = 225 (25.4%)		Immunotherapy, n = 463 (52.3%)		P-value
Drug treatment	During 1L period	>=1	884 (99.8)		176 (99.4)		225 (100.0)		463 (100.0)		<.001
		Median [IQR] drug lines per person per month	10.07 [6.32, 16.90]		13.26 [7.35, 18.84]		8.41 [4.44, 12.87]		10.60 [6.72, 16.89]		<.001
		Median [IQR] number of pharmacological groups per person per month	2.59 [1.23, 6.41]		4.49 [2.32, 8.37]		1.49 [0.85, 2.94]		2.59 [1.26, 5.88]		<.001
		Median [IQR] number of packs, per person per month	18.51 [10.68, 37.66]		21.29 [12.37, 45.34]		12.34 [7.22, 26.90]		19.74 [11.52, 39.44]		<.001
		>=1	886 (100.0)		177 (100.0)		225 (100.0)		463 (100.0)		NA
Drug treatment	During FU period	Median [IQR] drug lines per person per month	9.05 [5.77, 14.04]		10.61 [6.94, 15.96]		7.57 [4.29, 10.94]		9.59 [6.07, 14.56]		<.001
		Median [IQR] number of pharmacological groups per person per month	1.67 [0.92, 3.38]		2.07 [1.06, 3.88]		1.15 [0.72, 2.05]		1.79 [0.98, 3.67]		<.001
		Median [IQR] number of packs, per person per month	17.96 [10.14, 31.48]		22.81 [12.44, 35.70]		11.56 [6.55, 23.64]		19.18 [11.44, 33.79]		<.001
		>=1**	521 (93.9)		141 (95.3)		90 (90.0)		270 (94.1)		.218
		Median [IQR] drug lines per person per month**	12.00 [6.00, 21.00]		12.50 [7.00, 19.00]		10.00 [3.00, 20.25]		12.00 [7.00, 22.00]		.241
Hospitalization admissions***	During 1L period	Median [IQR] number of pharmacological groups per person per month**	7.00 [4.00, 12.00]		7.00 [5.00, 12.00]		7.00 [3.00, 11.25]		8.00 [4.50, 12.00]		0.461
		Median [IQR] number of packs, per person per month**	25.06 [10.12, 54.84]		25.07 [10.65, 52.21]		21.43 [5.00, 49.36]		26.75 [12.20, 57.34]		0.257
		>=1	316 (35.7)		43 (24.3)		78 (34.7)		189 (40.8)		0.001
		Median [IQR] number of admissions per person per month	0.21 [0.09, 0.48]		0.44 [0.20, 0.57]		0.13 [0.07, 0.23]		0.22 [0.10, 0.50]		<0.001
		Median [IQR] length of stay per person per month	0.71 [0.25, 1.88]		0.81 [0.43, 1.47]		0.55 [0.23, 1.06]		0.81 [0.23, 2.34]		0.077
Drug treatment	During FU period	>=1	676 (76.3)		151 (85.3)		145 (64.4)		362 (78.2)		<0.001
		Median [IQR] number of admissions per person per month	0.22 [0.10, 0.51]		0.27 [0.14, 0.54]		0.13 [0.07, 0.24]		0.25 [0.10, 0.55]		<0.001
		Median [IQR] length of stay per person per month	1.37 [0.42, 3.66]		1.81 [0.60, 4.17]		0.73 [0.22, 1.64]		1.68 [0.42, 4.12]		<0.001
		>=1**	368 (66.3)		108 (73.0)		68 (68.0)		182 (63.4)		0.089
		Median [IQR] number of admissions per person per month**	1.00 [1.00, 2.00]		1.00 [1.00, 2.00]		1.00 [1.00, 1.00]		1.00 [1.00, 2.00]		0.099
Drug treatment	Last month before death	Median [IQR] length of stay per person per month**	7.00 [4.00, 13.00]		7.50 [4.00, 13.00]		8.00 [3.00, 15.00]		7.00 [4.00, 13.00]		0.958

*Including patients that received any other treatment n = 21.

**For those that died, n = 555 (62.6% of patients died before end of study).

***For those hospitalized.

****CT, MRI, PET-CT.

Abbreviations: 1L, first-line; FU, follow up.

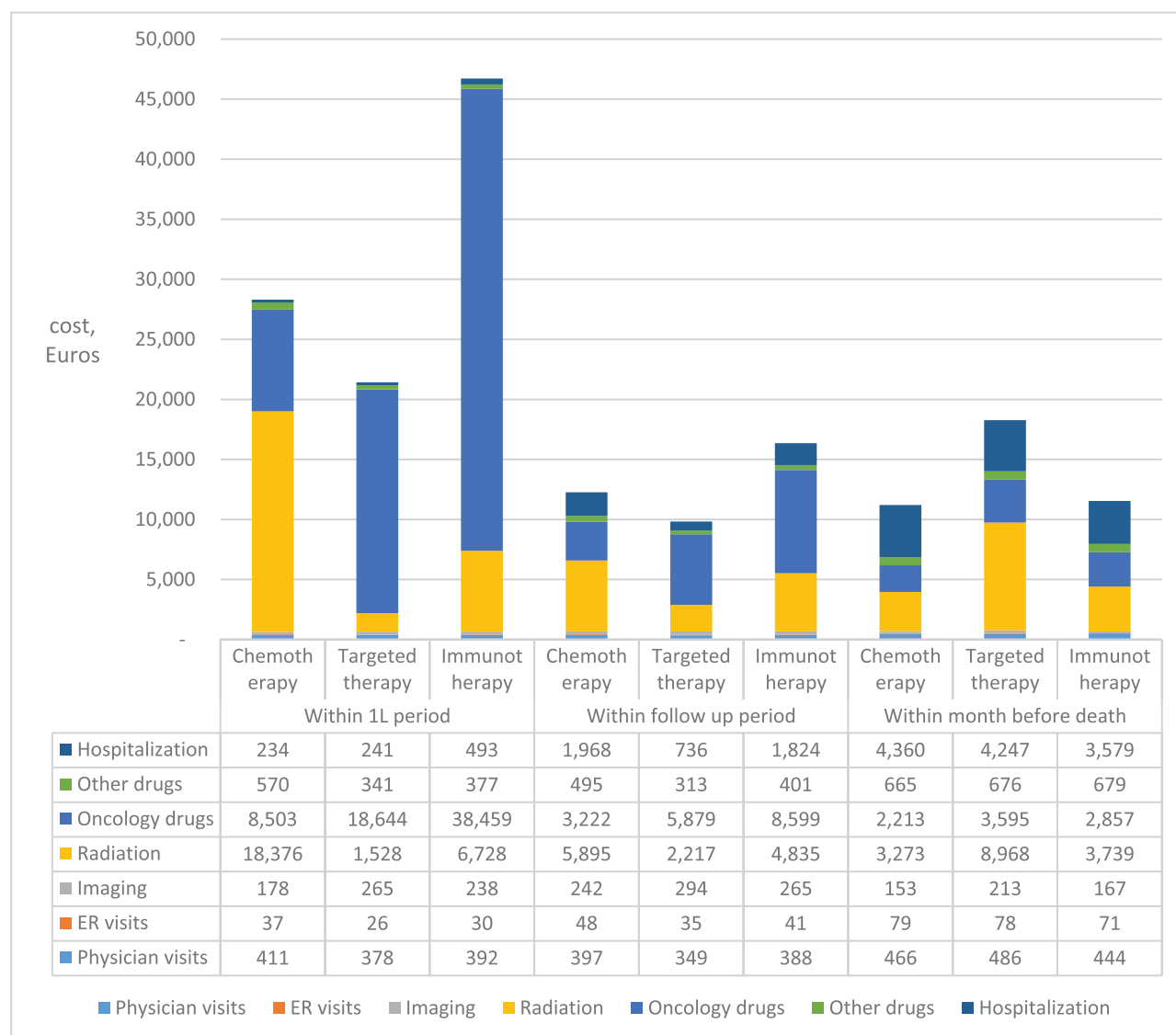


Figure 1. Healthcare resource utilization mean costs, per-person-per-month, in Euros, split by 1L treatment groups per treatment period.

Table 4. Mean total cost per patient for the entire follow up.

First-line treatment groups*	Number (%) died	Mean monthly cost per patient (PPPM) for entire follow up period (Euros)	Total mean cost per patient for those that died (Euros) N = 555
Platinum based chemotherapy	147 (83.05%)	12 267	121 154
Targeted therapy	100 (44.44%)	9823	129 458
Immunotherapy	287 (61.99%)	16 352	110 717

*Results for patients that received any other first-line treatment $n = 21$ are not shown.

All costs were based on the Israeli Ministry of Health Price List for Ambulatory and Hospitalization Services and prescription prices, which are maximum prices for these services. In practice, the healthcare providers can pay a lower price for services based on agreements.

The presented expenditures in the current study significantly underestimate the US costs as prices in Israel are lower per drug, and infusion costs were not included in any group. Still the expenditures were mostly dominated by medication

costs and were therefore compared between groups. The estimated comparison between US and other countries is summarized in a comparison of US and International Prices for Top Medicare Part B Drugs from 2018 and other studies.^{46,47}

We observed that total costs PPPM within the 1L period were mostly driven by drug costs for patients that received 1L targeted therapy and immunotherapy. Total costs for this period were highest for patients that received 1L immunotherapy due to high drug cost, followed by platinum-based

1L, where palliative radiotherapy contributed significantly as required by lack of more effective systemic treatment options.

Within the entire follow-up period, total costs were lower for all treatment groups as compared to the 1L period, probably due to decreased drug costs following cessation of 1L therapy, as most patients were treated with chemotherapy or best supportive care after disease progression on 1L therapy. Costs were more evenly spread between drug treatment and radiotherapy, and hospitalization costs increased as compared to the 1L period.

Radiation treatment was mostly in the palliative care setting and was not a combined treatment protocol. Due to an expected lower rate of benefit from systemic treatment, it was more common with platinum-based chemotherapy in 1L than with targeted therapy or immunotherapy. According to the national formulary, during 2017-2018 only patients with $\text{PD-L1} \geq 50\%$ could receive immunotherapy, and during 2019-2020 all patients could receive chemo-immunotherapy combination and the former group could still receive single-agent immunotherapy.

A separate analysis was conducted for patients who died and who therefore completed their follow-up period. A total of 83%, 62%, and 44% of patients died, for those who received 1L platinum-based chemotherapy, immunotherapy, and targeted therapy respectively. While monthly PPPM was highest for patients that received 1L immunotherapy followed by platinum-based chemotherapy, (where radiotherapy contributed significantly to cost), when evaluating the total follow-up period—mean cost was highest for those that received 1L targeted therapy, likely due to the prolonged treatment period and longer survival. As for clinical benefit, we observed that by the end of the study period, 55% of patients in the 1L targeted therapy group and 38% of patients in the 1L immunotherapy group were still alive, similar to expected results in other studies.⁴⁸

Findings from a systemic literature review of the economic impact of 1L immunotherapy in NSCLC showed that although costs of immunotherapy are higher than those of chemotherapy, increased treatment costs are balanced by lower HCRU and costs for adverse event management, including hospitalizations and emergency department visits.⁴⁹ These findings are corroborated in our study showing reduced total cost for the immunotherapy group.

In a study from 2012 to 2017 comparing 2L immunotherapy and chemotherapy, OS was longer for patients receiving immunotherapy. Hospitalizations were similar among the treatment groups, but more patients receiving immunotherapy had at least 1 adverse event, possibly due to the extended duration of drug exposure among these patients.⁵⁰

A previous study, that compared targeted therapy to chemotherapy among patients with EGFR-positive mNSCLC, found higher inpatient costs and inpatient days for 1L targeted therapy relative to chemotherapy which counters the assumption that patient quality of life would be improved. In that study, average total costs of care were higher in 1L targeted therapy patients than in patients receiving 1L chemotherapy.²⁶ These findings are corroborated in our study showing a higher cost for radiotherapy in the last month before death for the targeted therapy group.

Strengths and limitations

Strengths of this study include high-quality longitudinal data obtained from the MHS electronic database including all medical and billing data, and a comprehensive review

of patient medical records with a long follow-up. MHS comprises 25% of the patient population in Israel and shows real-world generalizability within the population.²⁹ Our study reflects current practice patterns and is the first study to report healthcare utilization and costs in Israel in the immunotherapy era. However, this study included only those patients who initiated 1L therapy and does not provide insight into untreated patients which according to our former study account for about 21% of patients.⁵¹ Our study does not include toxicity data and therefore does not fully describe the reason for treatment discontinuation, however, based on our daily work at the MHS medication pre-authorization department, the major cause for treatment change in metastatic patients is disease progression and/or death.

Limitations of our findings include the retrospective nature of this study and the change in health policy during the study period with the inclusion of chemo-immunotherapy from 2019. This is represented in the difference in median follow-up between the groups, and stems from the difference in date of medication inclusion in the National Medication Formulary as stated above, however, our cohort is clean and reliable. Smoking data were collected from physician reporting and classified into ever, never, or unknown so our data may not be accurate. Further studies are required to assess the impact of immunotherapy and other innovative treatments on the disease management costs of mNSCLC, including long-term outcomes for patients receiving immune checkpoint inhibitors.

Outpatient infusion billing was not included in our study because it is relatively cheap in Israel and expected to be similar per treatment month in the chemotherapy and immunotherapy groups.

Our data are therefore not complete but reflect the relations between groups.

Conclusion

This study reports baseline demographic characteristics, healthcare utilization, and economic burden of mNSCLC where 1L immunotherapy, chemotherapy, and targeted therapy were used to treat patients with mNSCLC in a real-world setting. While the significant cost in 1L of treatment was associated with patients who received 1L immunotherapy or targeted therapy due to high-drug costs, the mean total cost of those who died was higher in patients who initiated targeted therapy and platinum-based chemotherapy, likely due to palliative radiotherapy use in the latter group. The clinical benefit of targeted therapy is well established and reflects the unique tumor features, and the long 1L treatment period in those patients contributes overall to increasing the total cost.

To the best of our knowledge, we present the only HCRU and costs real-world study in Israel in a large cohort, with a long follow-up in the immunotherapy era. Although we report higher costs for patients who received newer drug treatments which are the main treatments for patients in 1L these days, expenditure was saved in other medical fields such as radiotherapy, and the patients gained longer survival, possibly a higher quality of life, and as for the immunotherapy treated patients a reduced cost compared to chemotherapy-treated patients.

This study can help healthcare organization decision makers and payers to better understand the cost expectancy for mNSCLC patients and better plan resource allocation.

Further studies are required to assess the impact of immunotherapy and other innovative treatments on the disease management costs of mNSCLC.

Author contributions

Conceptualization, methodology or analysis: Lior Apter, Sarah Sharman Moser, Gabriel Chodick, Nava Siegelmann-Danieli; writing original draft preparation Lior Apter, Sarah Sharman Moser, Nava Siegelmann-Danieli; final approval: Sarah Sharman Moser, Lior Apter, Gabriel Chodick, Nava Siegelmann-Danieli, Sivan Gazit, Moshe Hoshen, Dan Greenberg; Accountability: Sarah Sharman Moser, Lior Apter, Gabriel Chodick, Nava Siegelmann-Danieli, Sivan Gazit, Moshe Hoshen, Dan Greenberg; All authors have read and agreed to the published version of the manuscript.

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Conflicts of interest

All the authors declare that they have no conflicts of interest that might be relevant to the contents of this manuscript.

Ethical approval

The authors state that they have obtained appropriate institutional review board approval and have followed the principles outlined in the Declaration of Helsinki for all human experimental investigations. The study was approved by the local ethics review board of Maccabi Healthcare Services in Israel.

Consent to participate

Consent was waived by the ethics review board of Maccabi Healthcare Services in Israel due to the retrospective nature of this study.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due the Israeli Ministry of Health and MHS's data privacy policy but are available from the corresponding author on reasonable request.

Supplementary material

Supplementary material is available at *The Oncologist* online.

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